

Tutorial 2 Answer Key

1. 40 V
2. 6 As
3. $I_1 = -1.529 \text{ A}$, $I_2 = -0.676 \text{ A}$, $I_3 = -0.672 \text{ A}$
4. 0.3
5. -80
6. a) The problem with the node equation is that we cannot express the current in the branch between V_a and V_b , i.e., V_a and V_b are dependant variables.
b) 12.33 V
7. a) $V_G = V_T = V_{oc}$ & $R_L = R_G$
b) $R_L = R_G$
8. a) $A_v = \frac{-500 R_L}{50,000 + R_L}$
b) $A_v = -143$
9. Thevenin: $V_{th} = -84V$, $R_{th} = 20\Omega$
Norton: $I_N = -4.2A$, $R_N = 20 \Omega$
10. $-26.667 V$